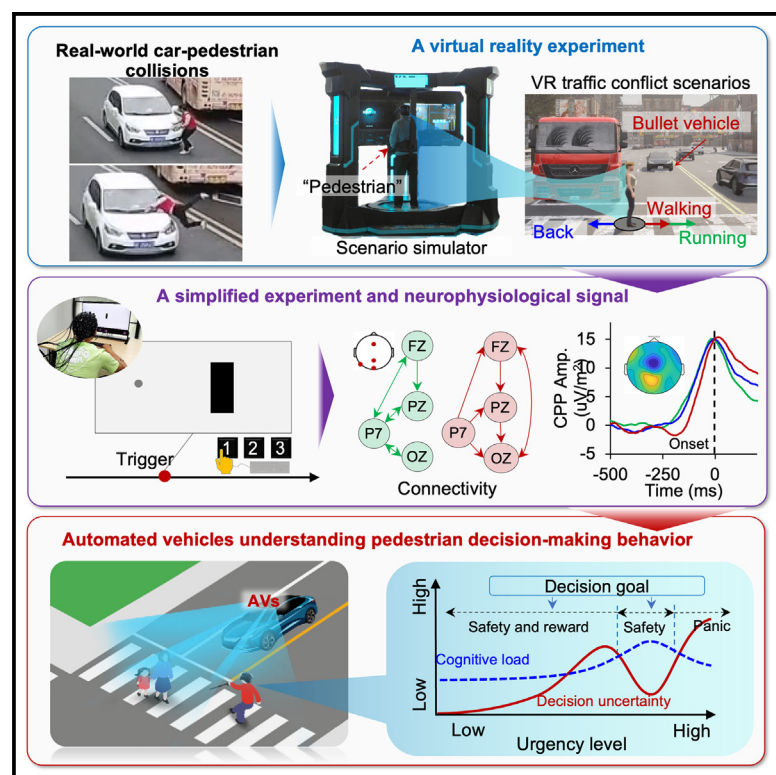


Pedestrian decision-making uncertainty in urgent scenarios modulates multi-level, neural hierarchies

Graphical abstract



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In brief

Li et al. investigate pedestrian decision-making uncertainty in urgent traffic scenarios, revealing a dynamic shift in goals: uncertainty initially decreases with rising urgency but then increases as urgency intensifies. Their findings can enhance autonomous vehicles' understanding of pedestrian behavior toward safer interactions.

Highlights

- Pedestrian decision-making uncertainty increases with urgency
- Neural activation in higher-order cognitive functions intensifies under increased urgency
- Automated vehicles enhance safety by understanding pedestrian decision-making



Article

Pedestrian decision-making uncertainty in urgent scenarios modulates multi-level, neural hierarchies

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SUMMARY

Uncertainty in pedestrian decision-making in urgent traffic scenarios significantly influences safe interactions with automated vehicles. However, the underlying mechanisms governing such decision-making behavior remain insufficiently understood. To bridge this gap, we design two experimental paradigms of varying complexity to simulate spatiotemporal pressure and the consequences of decision failures: a high-fidelity virtual reality pedestrian-vehicle interaction experiment and a simplified dynamic stimulus task. Our findings reveal that as stimulus urgency increases, humans adjust their decision-making goals, leading to an initial decrease in decision uncertainty, followed by an increase as urgency intensifies. Neurophysiological signal analysis suggests that insufficient perceptual information and evidence accumulation contribute to heightened decision uncertainty. Our study highlights a clear correlation between human decision-making uncertainty and scenario urgency, particularly within a defined urgency range. Integrating foundational cognitive principles and engineering approaches provides insights to address critical safety issues in pedestrian-vehicle interactions within automated transportation systems.

INTRODUCTION

Road traffic accidents remain a significant global public health concern, resulting in over 1.19 million fatalities and 50 million injuries annually.¹ With the increasing prevalence of automated vehicles (AVs) to enhance road safety, they must possess the capability to accurately interpret and foresee the actions of other road users to make effective decisions in safety-critical scenarios.² Failure to accurately predict and understand pedestrian behavior when encountering oncoming pedestrians can lead to severe traffic accidents, as illustrated by the first pedestrian fatality caused by an AV in Arizona, USA, in 2018. Pedestrians' uncertain behavior during traffic conflicts is a key factor affecting safety during these interactions^{3–5} (Figure 1A). For typical traffic scenarios, current research has extensively studied driver-pedestrian interactions using natural driving datasets, leading to the development of reliable predictive methods and models.^{6,7} However, in emergency traffic situations involving the risk of pedestrian collision injury, quantifying pedestrians' decision-making and subsequent motion poses challenges due to inherent limitations in the quality and quantity of accident data and investigation methods, particularly the lack of data on human reactions right before the event.^{8,9}

Understanding human behavior in road traffic scenarios, which includes drivers and vulnerable road users, requires recognizing their diverse communicative and cognitive processes.¹⁰ This un-

derstanding can enhance the design, functionality, and safety protocols of AVs by aligning technological advancements with human behavioral patterns and needs. In pedestrian-vehicle interactions, humans engage in specific traffic-related activities (e.g., crossing roads), perceive vehicle interactions as stimuli, and make decisions within specific time constraints. In autonomous vehicle research, there is often an emphasis on investigating how environmental factors, including speed, distance, and time to collision, impact pedestrian decision-making behaviors.^{11,12} To address this, researchers have developed a virtual reality (VR)-based pedestrian-vehicle interaction experimental method that simulates the full process of these interactions in a realistic manner.^{3,13,14} Such an approach effectively captures pedestrian decision-making behaviors in conflict traffic scenarios, providing a feasible and robust paradigm. However, such VR-based pedestrian experiments are limited to behavioral representations and lack deeper exploration of the underlying neural decision-making mechanisms—an area that remains underexplored in traditional traffic research.

Therefore, deeper exploration into the underlying mechanisms of human brain decision-making and theoretical modeling is warranted, but it is frequently overlooked in current studies. Cognitive neuroscience within the domain of statistical decision theory has developed a detailed framework that explains how decisions are formed by integrating prior knowledge, evidence, and value considerations to achieve predetermined goals.^{15–17} Such a



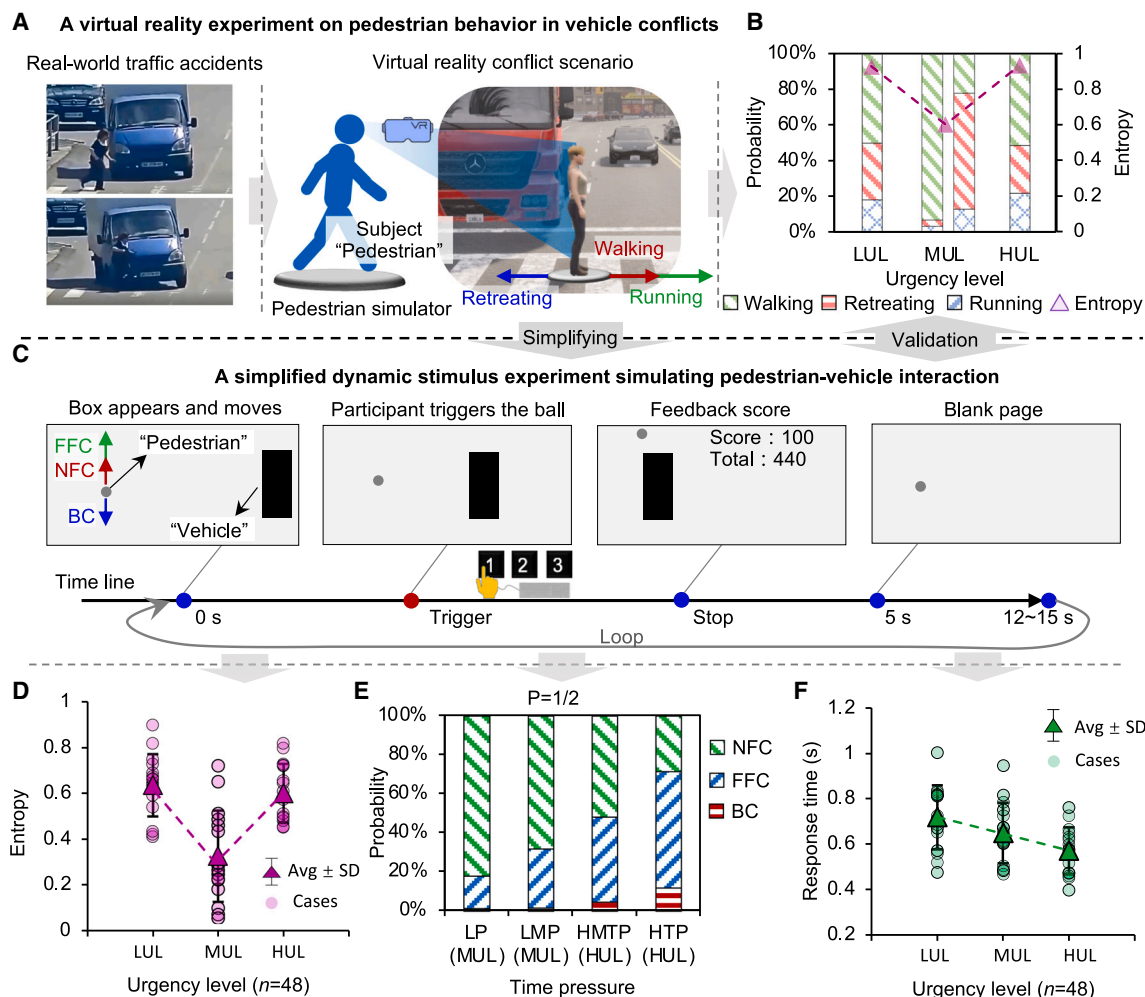


Figure 1. Research pathways, tasks, and behavioral data

(A) Left: diagram of vehicle and pedestrian conflict scenarios in the real world; right: illustration of virtual reality (VR)-based pedestrian-vehicle interaction experiment. Participants use a simulator and VR headset to enter a virtual environment. Depending on the traffic scenario, they choose to walk, run, or retreat to cross the road safely.

(B) Distribution of decision-making behavior of participants in a realistic VR experiment. HUL, MUL, and LUL denote high-, moderate-, and low-urgency levels, respectively. In the MUL category, the left bar represents regions close to the vehicle's centerline with higher time-to-collision (TTC) values (top left of Figure S2), while the right bar represents regions farther from the centerline with lower TTC values (bottom right of Figure S2). Therefore, the decision uncertainty entropy in the MUL category is the mean entropy of this region.

(C) A simplified dynamic stimulus experiment: participants controlled the gray dot to move to avoid collision by approaching the black rectangle using a physical button. Decision categories of participants include NFC with the highest reward (key 1), FFC with the second-highest reward (key 2), and BC without a reward (key 3). Participants were rewarded or punished with the points they gained in each trial. In the case where the decision-making results in a collision, a high penalty is applied, leading to the clearing of the total score.

(D and F) The impact of urgency level on response time and decision-making uncertainty. *n* represents the sample size.

(E) The probability distribution of the three decision-making behaviors under various time pressures at the initial relative position between the point and the rectangular block ($P = 1/2$).

framework covers crucial parameters, including evidence accumulation and decision variables, to define the mechanisms and thresholds associated with value-based decision-making.¹⁸ Moreover, experimental studies have identified neurophysiological signal features, such as movement-related potentials (MRPs), centro-parietal positivity (CPP), and connectivity indices, which are associated with decision-making processes.^{19–21} These features reflect the accumulation of evidence for decision

variables and the involvement of top-down/bottom-up attention mechanisms. This alignment establishes a recognized neural mechanism foundation for theoretical models, such as decision-related drift-diffusion models.^{22,23} Current research on neural mechanisms has primarily relied on simple experiments analyzing the monotonic effects of single-factor variables on neural activation states.²⁴ These studies offer a theoretical basis and classic experimental paradigms for further exploration into the

uncertainty in decision-making associated with stimulus variables, neural activation, and decision feedback in real-world interactions.

With a focus on decision-making mechanisms, particularly in urgent scenarios, current research suggests that in response to urgency stimuli induced by speed or time pressure, the baseline, rate, and threshold of evidence accumulation can dynamically adapt to prioritize either speed or accuracy.²⁵ In high-urgency situations characterized by increased time pressure, the baseline of action-triggering signals increases during the static phase before the stimulus onset and then rises rapidly after the stimulus onset.^{24,26} This adaptive adjustment reduces the required evidence accumulation, facilitating swift and accurate decision-making. Furthermore, the intricate nature of human decision-making often involves extensive adjustments across multiple neural hierarchies. Consequently, neurophysiological research has also explored the role of broader neural modulation systems in the decision-making process.^{27,28} Under urgency conditions, distinct sensorimotor cortical areas exhibit differential influences, wherein early perceptual encoding is enhanced to some extent, mitigating the impact of motor-level urgency on decision accuracy.^{27–29} Existing research has primarily focused on the trade-off mechanism between speed and accuracy. Investigations into behavioral uncertainty in making accurate choices, such as understanding pedestrian behaviors in interactions with vehicles, have not been covered. Additionally, the mechanisms of encoding decision evidence information under the combined influence of factors like speed pressure, spatial pressure, punitive consequences of decision failures, and other contextual elements remain unclear.

In this study, we conducted two levels of experiments with varying complexity: a VR-based pedestrian-vehicle interaction experiment (VR experiment) and a simplified dynamic stimulus experiment (simplified experiment). In the VR experiment, we created realistic virtual traffic scenarios where participants, equipped with VR headsets, entered a simulated traffic scenario to interact with conflicting vehicles. Conversely, the simplified experiment focused on critical elements extracted from real traffic scenarios, emphasizing large-scale experimental data and extracting multidimensional physiological signals. Specifically, the simplified experiment introduced urgency by incorporating temporal and spatial pressures associated with stimulus input. The simplified experiment was conducted in a laboratory, where physiological signals like electroencephalograms (EEGs) (Figure S1) and eye movements were considered feasible and usable. Such effective signals were analyzed to quantify the mechanism of uncertainty in pedestrian decision-making.

Our results suggest that decision uncertainty varies with increasing urgency, potentially due to the dynamic adaption of decision goals. Moreover, urgency directly influences perception processes and enhances neural activation in higher-level cognition functions. For instance, features of frontocentral negativity (FCN) and MRPs related to motion preparation and of CPP related to evidence accumulation all exhibit a negative correlation with decision uncertainty across various urgency levels. Nevertheless, as the urgency level increases, top-down cognitive mechanisms are engaged at an earlier stage. Finally, we pro-

pose a model framework that describes the variation of decision uncertainty with increasing urgency. The conclusions are anticipated to improve the predictability of pedestrian behavior in scenarios when AVs face conflicts and guide the vehicles in making safe avoidance decisions.

RESULTS

Decision uncertainty in pedestrians under urgent scenarios

We employed VR technology to simulate realistic traffic conflict scenarios by manipulating the sudden appearance time and relative position of vehicles,³ thereby creating near-real environmental stimuli of varying urgency for pedestrians (participants) crossing the street (see Figure 1A and the methods section for details). To validate the simplified experiments effectively, we accounted for the range of pedestrians' physical characteristics and the time delay from perception to movement. We performed a comparative analysis on cases where $1.0\text{ s} < \text{TTC} < 3.0\text{ s}$ and the pedestrian's offset from the vehicle's centerline was between 0.5 and 2.0 m (Figures 7C and S2B). The results indicate that pedestrians tend to choose walking when they are closer to the vehicle's centerline and the TTC is larger (top left region of Figure S2B), whereas they tend to choose to retreat when they are farther from the vehicle's centerline and the TTC is smaller (bottom right region of Figure S2B). Furthermore, the entropy characterizing the uncertainty of pedestrian decisions shows that as urgency increases from a low-urgency level (LUL) to a high-urgency level (HUL), decision uncertainty first decreases and then increases (Figure 1B).

To collect large-scale experimental data and high-quality physiological signals during pedestrian decision-making, we employed a simplified experiment designed to simulate pedestrian-vehicle interactions. This experiment was inspired by classic collision judgment paradigms³⁰ and integrated essential variables representing pedestrian-vehicle dynamic interactions³ (see Figure 1C and the methods section for details). The recorded data encompassed response time, decision-making categories, decision-making uncertainty, EEG signals, and eye movement. Response time was defined as the duration from the initiation of a stimulus to the participant pressing the key. The decision categories were classified into three distinct types: normal velocity forward crossing (NFC), which corresponds to walking; fast forward crossing (FFC), which corresponds to running; and backward crossing (BC), which corresponds to retreating. Participants were instructed to select an avoidance behavior from a set of three available options to maximize benefits while preventing collisions to the best of their ability. To analyze the influence of urgency on decision uncertainty, we conducted statistical comparisons and found a significant negative correlation ($F(2, 52) = 19.106, p < 0.001$) (see Figure 1D).

Participants showed the least decision uncertainty in scenarios with moderate-urgency levels (MULs). A consistent trend is observed both in the simplified and VR experiments (Figure 1B vs. Figure 1D). Furthermore, a two-way ANOVA of the simplified experiment results indicated that both time pressure and initial position significantly influence decision uncertainty ($F(6, 26) = 7.632, p < 0.001$) (Figure S3A). Specifically, when facing

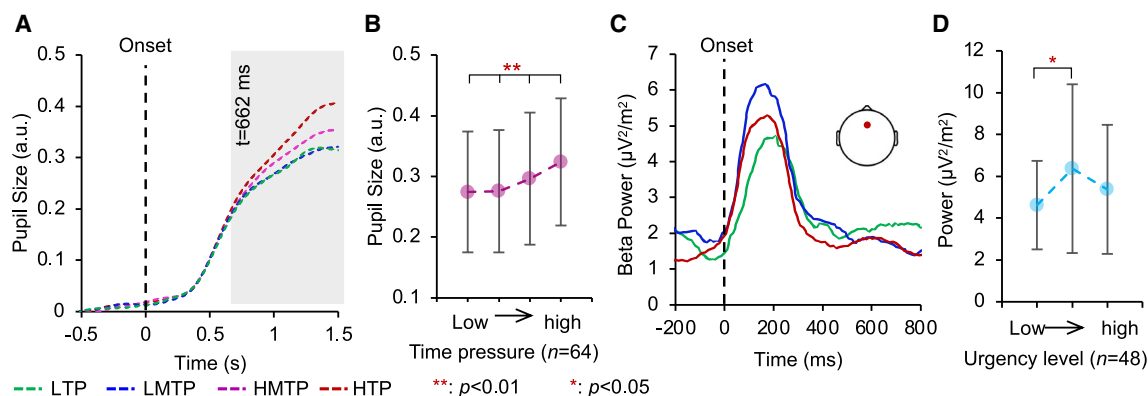


Figure 2. The change of pupil size and EEG beta wave with urgency levels

(A) The change of average pupil size under different time pressures. Shaded gray illustrates the time frame during which the differential across stimulus levels was significant (ANOVA: $p < 0.05$). The high, high-moderate, low-moderate, and low time pressures are denoted as HTP, HMTP, LMTP, and LTP, respectively. (B) The mean and standard deviation of pupil size from when significant differences were observed up to 1,500 ms. (C) The average curve of the power amplitude of the beta wave (13–30 Hz) at the frontal zone (FZ) electrodes. (D) The mean and standard deviation of the power amplitude peak under different urgency levels. Statistical significance for pupil size and power amplitude peak was determined using the Friedman test (F), with asterisks representing the significance levels.

increased time pressure at the same initial position, participants prioritized safer behaviors over those with higher potential rewards (Figures S3B–S3D). For instance, when positioned at $P = 1/2$ (where the collision risk is identical for NFC with a high reward and BC with no reward and lowest for FFC with the second-highest reward), escalating time pressure resulted in a higher frequency of both FFC and BC choices (see Figure 1E). In the VR experiment, pedestrian decision-making follows a consistent trend, as illustrated in Figure S2.

Specifically, with the same offset displacement, participants are less likely to choose walking and more likely to choose retreating or running as the TTC decreases. This suggests that, as urgency increases, participants tend to favor the safer options of FFC (running) and BC (retreating) over the higher-reward option of NFC (walking). Moreover, under high-urgency conditions, panic-induced confusion may lead participants to select the BC (retreating) behavior. The distribution of decision-making behavior and decision uncertainty exhibits the same trend in both simplified and VR experimental results.

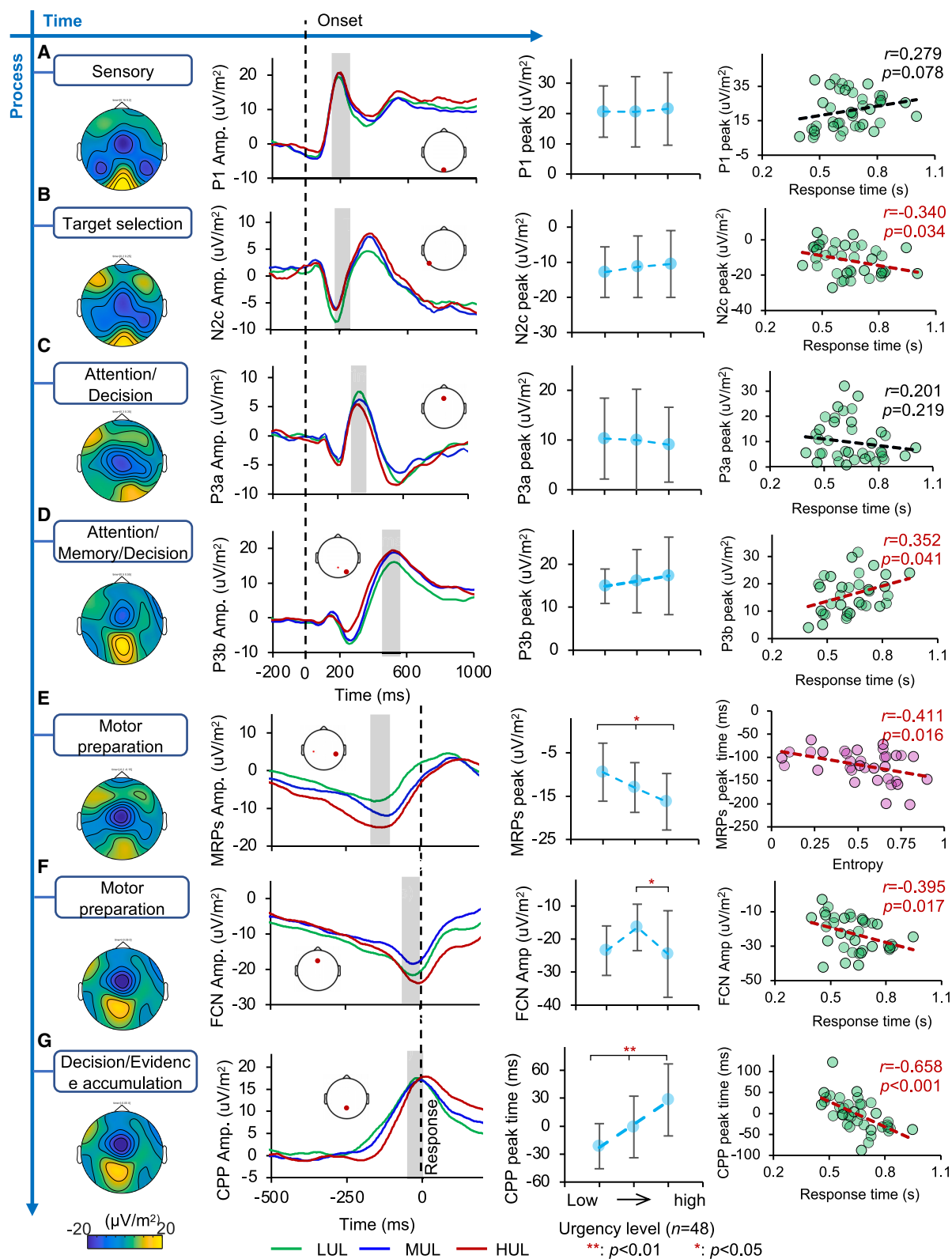
Based on the comparison and analysis of the data between the two sets of experiments, the results of the simplified experiment effectively reflect the decision-making uncertainty of pedestrians in near-real emergency scenarios. Consequently, we hypothesize that it stimulates similar decision-making processes in the human brain. Based on this premise, the following sections will analyze and discuss the behavioral data and physiological signal characteristics collected from the simplified experiment. This analysis aims to explore the impact of urgency on multi-level neural processes and identify neural and physiological markers associated with decision uncertainty.

Furthermore, the simplified reaction time behavior in the experiment shows a significant correlation with the stimulus conditions. We examined the impact on response times by treating time pressure and spatial constraints as a unified variable and then calculated the Pearson correlation coefficient (r) and performed a two-tailed test to determine the consistency of varia-

tion trends in response times between the stimulus parameters (see the methods section). The results suggest that the majority of participants (group A) primarily rely on perceiving spatial location for rapid sensing and decision-making. In contrast, a minority of participants (group B) rely on perceiving the speed of dynamic stimuli to make faster decisions (Figures S4 and S5). Moreover, one-way ANOVA analysis revealed that urgency levels, defined based on initial position and time pressure as illustrated in Figures 6 and S4, significantly affected the response times of both group A and group B (ANOVA: $F(2, 64) = 12.986$, $p < 0.001$, Pearson: $r = -0.347$, $p < 0.001$). The combined analysis of data from groups A and B revealed a significant decrease in response time with increasing urgency (Friedman: $F(2, 16) = 28.500$, $p < 0.001$; Pearson: $r = -0.440$, $p < 0.002$) (see Figure 1C). This suggests that participants achieve faster response times under higher urgency by integrating perceived spatial and speed information.

Pupil size and EEG beta-band power increase with urgency

Related theoretical work suggests a correlation between perception-related cognitive effort and pupil size under time pressure. The pupil-related arousal system may generate a sense of urgency.^{28,31} We explored the effects of time pressure and urgency (spatiotemporal pressure) on pupil size and observed a sustained increase after the stimulus onset. Notably, this increase was more pronounced in amplitude and growth rate under high time pressure and urgency, as illustrated in Figures 2 and S6. We further used one-way ANOVA (two-tailed) on pupillometric data within a 200 ms time window, using a 3 ms sliding time step, to identify the time points at which stimulus levels significantly influenced pupil size. The results indicated that time pressure significantly impacted pupil size, particularly evident at 662 ms after stimulus onset (ANOVA: within a 662 ± 100 ms time window, $F(3, 60) = 2.827$, $p = 0.039$, as illustrated in Figure 2A). Additionally, we calculated the mean and SD of



(legend on next page)

pupil size from the time points when significant differences were observed up to 1,500 ms and conducted a correlation analysis, which revealed a significant positive correlation between time pressure and pupil size (Friedman: $F(3, 16) = 12.375, p = 0.006$; Figure 2B), aligning with Steinemann et al.'s findings.²⁹ The impact of urgency level on pupil size is insignificant (Friedman: $F(2, 16) = 1.125, p = 0.570$; Figure S6).

Previous relevant research indicates that increased beta-band (13–30 Hz) activity might reflect higher tension, vigilance, and cognitive load in participants.^{32,33} The power of the alpha-band peaks within approximately 200 ms of stimulus onset and then declines (see Figure 2C). We then compared the powers of the beta band at the frontal electrode across different urgency levels. The analysis revealed a significant effect of urgency level on the amplitude peak of the beta band (Friedman: $F(2, 16) = 6.533, p = 0.038$), with the highest power peak observed at a MUL. This finding suggests that participants may experience a higher cognitive load during decision-making under stimuli with moderate urgency.

Correlation of late-stage EEG features with urgency

To investigate the comprehensive neural activation hierarchy throughout the decision-making process, various neural metrics derived from EEG signals associated with perception-decision execution were assessed (see Figure 3). Using scalp EEGs, differences in neural physiological markers were tracked, including the P1³⁴ (sensory information encoding) in the primary visual cortex, the N2c³⁵ (target selection) in the inferior temporal cortex, the P3a³⁶ (attention and decision) in the prefrontal cortex, the P3b³⁷ (attention, memory, and decision-making) and CPP^{38,39} (evidence accumulation and decision-making) in the central parietal cortex, the frontal preparatory activity (FCN),^{40,41} and the MRPs⁴² (motor preparation) in the motor cortex, across various stimulus conditions. The sequence of brain region activation throughout the decision-making task timeline was first determined. Subsequently, an analysis was conducted within this timeline, examining the peak amplitudes and latency of P1, N2c, P3a, and P3b, as well as the pre-response amplitude, build-up rate, peak amplitudes, and peak timing (relative to the response time) of MRPs, CPP, and FCN. The significance and correlation of urgency's impact on all features were assessed using the Friedman test and the Spearman correlation coefficient.

In the stimulus-locked analysis, the EEG activity shows that the urgency level does not significantly affect the P1, N2c, P3a, or P3b components. However, the Spearman correlation analysis indicates a significant negative correlation between the N2c amplitude peak and response time, suggesting that faster responses associated with attention require a higher resource load

(Spearman: N2c peak vs. response time: $r = -0.340, p = 0.034$). Conversely, a significant positive correlation exists between P3b amplitude peak and response time, indicating that fast responses may lead to a decrease in the activation level of late-stage functions associated with advanced decision-making (Spearman: P3b peak vs. response time: $r = 0.352, p = 0.041$; see Figures S7–S10 for details).

The response-locked analysis revealed that the urgency level significantly impacts various components of event-related potentials. Specifically, the Friedman test indicates that the urgency level has a significant impact on the amplitude peak of MRPs ($F(2, 16) = 7.000, p = 0.030$), the peak timing of CPP ($F(2, 16) = 10.255, p = 0.006$), the build-up rate of CPP ($F(2, 16) = 6.167, p = 0.046$), the amplitude peak of FCN (medium urgency vs. high urgency: $F(1, 16) = 4.000, p = 0.006$; medium urgency vs. low urgency: $F(1, 16) = 10.000, p = 0.023$), the peak timing of FCN ($F(2, 16) = 8.000, p = 0.018$), and the pre-response amplitude of FCN (medium urgency vs. high urgency: $F(1, 16) = 10.000, p = 0.023$).

Further analysis revealed that the peak amplitude of MRPs and the build-up rate of CPP increase with higher urgency levels. The peak times of CPP and FCN are delayed as the urgency level increases. Notably, the peak of CPP occurred before the response in LULs and MULs and after the response in HULs. Similarly, the peak of FCN occurred before the response in all urgency levels and earlier than the CPP peak time. These findings suggest that HULs prompt participants to make rapid decisions even before evidence accumulation reaches its peak, reducing CPP amplitude as response time decreases. The results from the Spearman correlation analysis indicate a significant negative linear correlation between the peak timing of MRPs and decision-making entropy ($r = -0.411, p = 0.016$), while the peak timing of CPP shows a significant negative correlation with response time ($r = -0.658, p < 0.001$). Additionally, there was a positive correlation between CPP pre-response amplitude and response time ($r = 0.408, p = 0.025$), while a negative correlation existed between FCN pre-response amplitude and response time ($r = -0.395, p = 0.017$; see Figures S11–S13 for details).

Top-down control strategies in early perception processes

We used the Granger causality test to analyze the causal interactions among different cerebral regions' activation during decision-making. We calculated correlation coefficients for electrodes at frontal zone (FZ), parietal zone (PZ), occipital zone (OZ), and P7 in 100 ms windows from 0 to 700 ms (see Figure 4).

Under low-urgency conditions, the brain's information processing commences within 100 ms of stimulus onset, starting from the primary visual cortex (OZ) in the occipital lobe and progressing to

Figure 3. EEG characteristics of the process of decision-making

(A–G) Characteristics of EEG signals during the "perception-decision-execution" stages. These include EEG topography, the dynamic changes in EEG components, the relationship between EEG components and urgency, and the relationship between EEG components and response time. Specifically, the first column (from left to right) of graphs presents topographical maps of EEG activity at different stages of information processing, with each topographical map corresponding to the time interval indicated by the gray shaded area in the time-series plots of EEG components in the second column. The second column of graphs displays time-series plots of EEG signal features under different levels of urgency stimuli. The third column displays the mean and standard deviation of EEG signal features under three urgency levels. The fourth column represents the distribution of behavioral data for response time and decision uncertainty. Statistical significance of EEG signal features was assessed using the Friedman test (F), with asterisks denoting different significance levels.

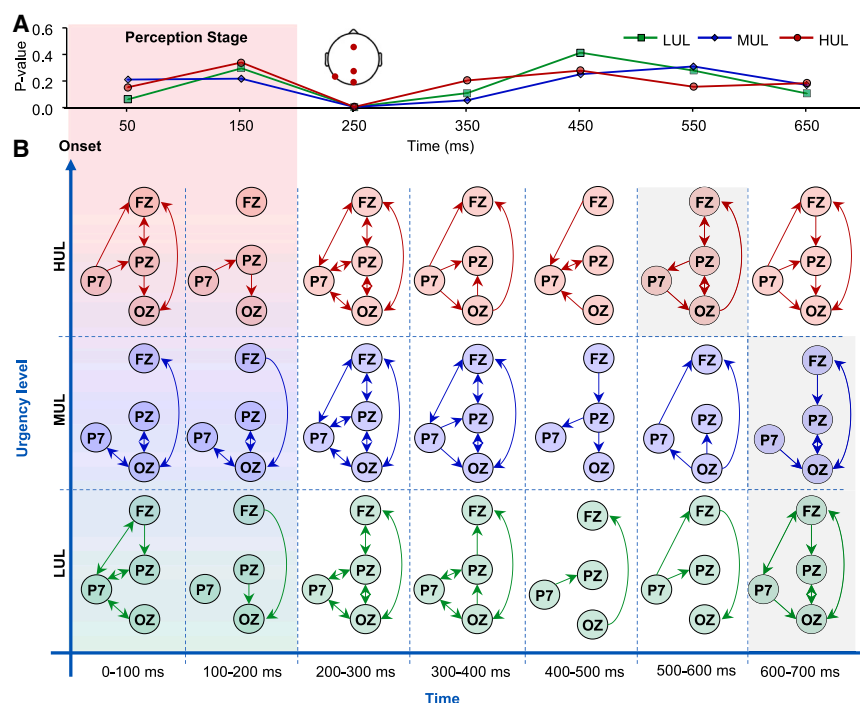


Figure 4. The connectivity among FZ, PZ, P7, and OZ electrodes based on Granger causality analysis

(A) The mean values of Granger causality significance p values across the four electrode signals under three levels of urgency. A lower p value indicates higher connectivity between brain regions.

(B) The arrows in the representation indicate the direction of information transmission among four electrodes, with significant impact ($p < 0.05$). The gray area represents the interval within which the average response times are located.

the target selection cortex (P7) in the temporal lobe. Subsequently, the processing proceeds to more intricate levels in the frontal (FZ) and parietal lobes (PZ). However, when moderate- and high-urgency conditions are present, the activation of the temporal and occipital lobes has a reduced impact on the activation states of the frontal and parietal lobes. In contrast, the activation of the frontal and parietal lobes directly influences the activation of the occipital lobe. This indicates a decrease in bottom-up cognitive processes with escalating urgency levels accompanied by an augmentation of top-down cognitive processes.²¹

Between 100 and 200 ms, there is a reduction in information flow across the entire brain domain. The activation of the frontal and parietal lobes significantly amplifies their impact on the activation of the occipital and temporal lobes, indicating a distinct top-down cognitive process. Subsequently, between 200 and 400 ms, there is an increase in neural activity across various brain levels, characterized by nearly all electrodes participating in mutual information transfer. This finding indicates the brain's active integration of multiple pieces of information to facilitate higher-level cognition and decision-making, with no significant differences observed across three urgency levels.

Between 400 and 700 ms, connectivity among brain regions decreases, potentially due to the reallocation of brain resources toward motor control following the completion of perceptual decision-making. Alternatively, it is posited that the brain undergoes a secondary round of information processing. In low-urgency conditions, this involves a second round of initial bottom-up processing followed by a top-down cognitive process, culminating in a response action at around 721 ms (mean response time). Under moderate-urgency conditions, the brain starts with top-down information transfer, followed by bottom-up transfer, leading to response actions occurring at approxi-

mately 655 ms. Conversely, in high-urgency conditions, the brain commences with top-down information transfer, followed by the activation of multiple neural levels, resulting in response actions executed at approximately 575 ms.

DISCUSSION

Daily, each individual may act as a pedestrian on the road, encountering typical pedestrian-vehicle conflicts, often accom-

panied by uncertainties. We validated that the decision-making patterns observed in the simplified dynamic stimulus experiment accurately reflect pedestrian behavior in real-life emergency traffic scenarios through a realistic VR experiment. Urgency levels directly affect decision-making uncertainty, revealing a multi-level neural activation process adjusted by neurophysiological indicators. These adjustment processes span the entire brain domain, involving the integration and processing of information. They are intricately linked to the level of neural activation, shedding light on the relationship between brain region synergy and resource allocation. Human decisions, in turn, result in subsequent motions during the interaction. Consequently, this study delves into the potential application of these findings in a different context: the decisions made by AVs, which play a crucial role in ensuring safety and balancing efficiency.

Under the urgency stimuli designed in this study, decision uncertainty initially decreases and then rises with increasing urgency. We propose a model framework, which is a simplified version of the pedestrian-vehicle interaction, to illustrate the relationship between urgency and decision uncertainty. It explains that the variation in human decision uncertainty across different urgency levels is associated with their decision goals and is influenced by cognitive load and evidence accumulation during decision-making (see Figure 5). Decision goals prioritize safety and reward maximization in low-urgency scenarios or scenarios with almost no urgency (e.g., in the VR experiment where TTC > 4.0 s and offset distance > 2.5 m). Humans require a lower cognitive load or less evidential information for decision-making. The need for cognitive load and evidential information in decision-making increases as urgency increases (e.g., in the LUL state in both the VR and simplified experiments). Faced with complex decision goals, the limitations of decision resources

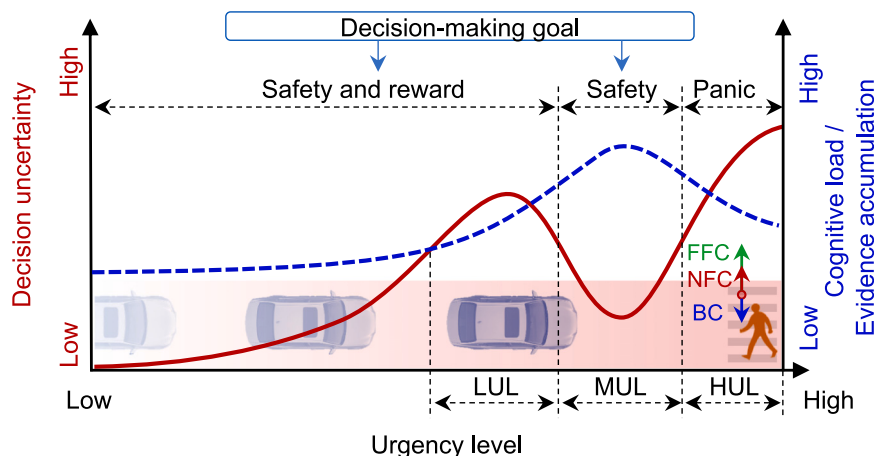


Figure 5. The model of decision-making uncertainty

The red solid curve represents decision uncertainty, while the blue dashed curve depicts changes in decision confidence and cognitive load as urgency levels increase. The proposed model framework can be explained by analyzing neurophysiological signals from different aspects: the stress or cognitive load, the evidence accumulation, and the motion preparation. Firstly, beta-band power amplitude, as a marker associated with stress or cognitive load, is highest under moderate urgency. Specifically, spectral analysis of EEG signals revealed a notable increase in power within the beta band, indicating a heightened state of cognitive load or induced anxiety as urgency increases among participants. Such an observation is consistent with prior theoretical research.^{32,33} Interestingly, the variation trend of

beta-band power amplitude with urgency level aligns with the pattern of changes in decision uncertainty (see Figures 1F and 2D), suggesting a probable association between increased cognitive load and increasing urgency levels. When urgency surpasses a certain threshold, additional factors such as heightened stress or constraints on cognitive time seem to play a role in reducing cognitive load.

contribute to increased decision uncertainty. As urgency continues to increase and decision goals are simplified to prioritize safety (e.g., entering the MUL state), the requirement for decision resources reduces, resulting in decreased decision uncertainty (“pedestrian confidently crosses the street”). However, if urgency further increases (e.g., entering the HUL state), then a reversal occurs, as heightened stress induced by urgency and limited cognitive time contributes to increased decision uncertainty, leading to a panicked pedestrian.

Secondly, our findings indicate that urgency does not directly influence CPP pre-response amplitude and build-up rate, but it does significantly impact the peak time of CPP, leading to a shift toward later as the urgency increases (see Figure 3G). The evidence accumulation represented by CPP can,⁴⁰ to some extent, explain the trend of decision uncertainty caused by changes in decision goals. In low-urgency scenarios, although CPP reaches its peak before decision execution, decision uncertainty remains high. This can be attributed to the interference from the goal of maximizing utility in low-urgency scenarios, which disrupts the stability of rapid decision-making. In contrast, in moderate-urgency scenarios, prioritizing safety weakens the pursuit of maximizing benefits, resulting in more stable decision-making. Previous research has highlighted the impact of urgency and evidence accumulation on decision accuracy and speed.³¹ Our findings further suggest that decision uncertainty is related not only to the threshold of evidence accumulation but also to the moment when evidence accumulation reaches its peak.

Thirdly, the MRP amplitude significantly increases with increasing urgency (see Figure 3E). This aligns with early research findings indicating that urgency (speed pressure), independent of evidence, affects MRP signals and downstream muscle activation without directly impacting upstream levels.²⁹ Conversely, the later-occurring FCN amplitude related to movement preparation initially increases with growing urgency and then decreases, aligning with the trend of decision uncertainty (see Figures 1D and 3F). FCN is more frequently observed and measured in experiments involving decision feedback, and

research results indicate a specific correlation between its post-decision EEG markers and the accuracy or confidence of decision-making.⁴² FCN not only is associated with motor preparation but may also represent the accuracy or confidence level of decision behavior. This also implies that decision uncertainty is linked not only to the decision-making process itself but to the motor control of behavior.^{28,29} These findings explain the variation in decision uncertainty with changes in urgency from the action preparation potentials.

Finally, urgency enhances overall information exchange in the brain, particularly reinforcing direct information transmission from the frontal lobe to the occipital lobe (Figure 4), despite the impact not being significant on EEG components during the perception phase. This augmentation indicates a more direct top-down perceptual process in the initial stages of perception, operating to expedite cognition and decision-making.⁴³ Specifically, as urgency increases, the brain may rely more on prior experience for direct top-down information integration, enabling rapid decision execution. This phenomenon could account for the elevated decision uncertainty in high-urgency scenarios, due to insufficient perceptual information availability. This result explains the lower accumulation of evidence associated with the reaction time under HULs mentioned earlier. Similarly, under moderate urgency, humans mitigate their demand for informational resources by simplifying decision goals, reducing decision uncertainty.

The question of how to interact with a moving pedestrian stands as one of the most frequently asked questions of AVs. The fundamental mechanism behind this inquiry is rooted in the inherent uncertainty in human decision-making, particularly under the combined influence of decision goals, rewards, limited time, and beyond. In scenarios with lower urgency than LULs, rich pedestrian-vehicle interaction data can be used to calibrate pedestrian behavior models. Models that consider traffic gaps and dynamic interaction times can accurately predict pedestrian behavior, indicating high predictability in pedestrian decision-making.^{44,45} In high-risk urgency scenarios involving collisions,

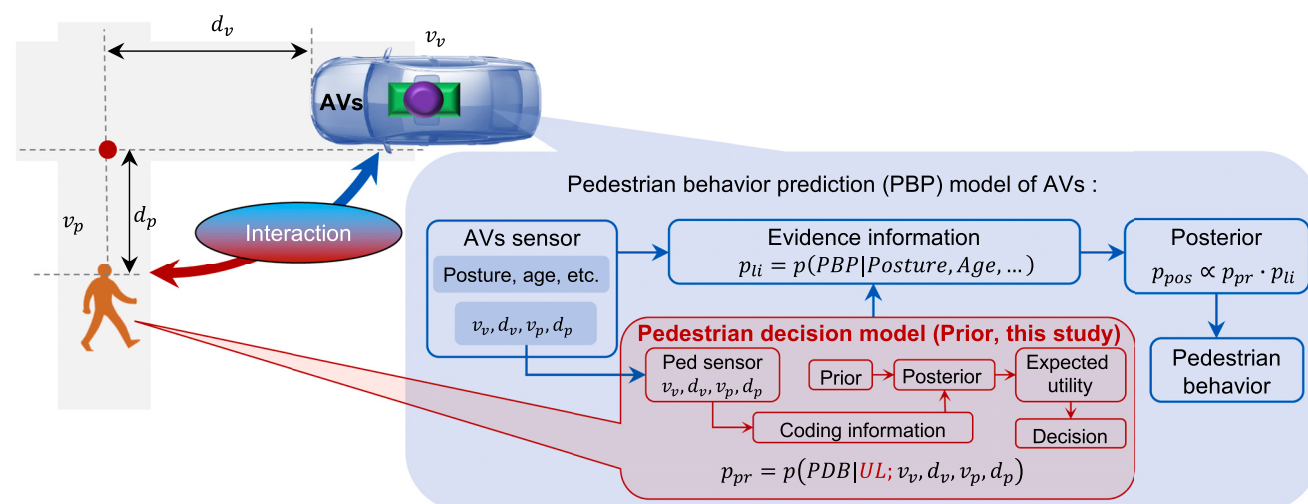


Figure 6. The potential application of analyzing pedestrian decision-making mechanisms in establishing pedestrian behavior prediction models for AVs

The pedestrian behavior prediction model consists of two nested Bayesian inference models. The first layer (blue flowchart) uses the perception information of automated vehicles (AVs) as evidence, while the prior information of AVs serves as the pedestrian decision-making model (second layer, red flowchart), which is the potential application object of this study. The pedestrian decision-making model describes how pedestrians encode information (evidence), decision uncertainty in different emergency scenarios (prior), and the balance between potential benefits and risks of crossing the road in different emergency scenarios (expected utility).

the distribution of uncertainty in pedestrian decision-making closely aligns with the present experimental findings, exhibiting variations across different levels of urgency (see Figure 5). Understanding and, more importantly, adapting to this inherent nature is crucial for AVs to navigate dynamic scenarios and ensure safe, efficient interactions with pedestrians.^{2,46} For instance, when pedestrian behavior falls within a pre-defined range of urgency, implementing actions such as braking, steering, or maintaining normal travel based on the pedestrian's movement becomes feasible and efficient. However, when urgency exceeds this range and pedestrian decision-making uncertainty increases, predicting their actions becomes challenging. In such cases, vehicles must adopt a more conservative strategy, such as applying only braking for deceleration to ensure safety.

Currently, AVs typically rely on pedestrian historical trajectories for predicting pedestrian behavior.⁴⁷ However, in real traffic scenarios, pedestrians exhibit nonlinear behavior that can change instantaneously based on surrounding environmental changes. In future applications of autonomous vehicles, a significant application of our present findings is the potential utilization in constructing a *priori* pedestrian decision-making models within the Bayesian inference framework (Figure 6). This application can greatly enhance AVs' ability to interact with pedestrians, which provides AVs with a *priori* knowledge of potential pedestrian decision-making in various scenarios—knowledge currently lacking in application. For example, we hope to establish behavioral prior models for pedestrian decision-making through a detailed analysis of the mechanisms revealed in this study. Such a model covers developing the information accumulation process during the perception stage and the variability in decision-making thresholds (CPP feature, Figure 3) within the Bayesian inference model framework. Further-

more, expected utility models can be created to consider the correspondence between urgency (manifested through speed, relative distance, etc.) and decision-making behavior (Figure 5). In that way, the model, anticipated to contribute to future AVs, addresses high-urgency hazardous scenarios often absent in natural driving datasets, while accommodating the inherent uncertainty in pedestrians' decisions during urgent situations.

In summary, our findings elucidate the mechanisms and patterns of uncertainty in human decision-making behavior under urgent, dynamic situations. We identified key trends in decision uncertainty that have significant implications for enhancing safety in human-machine interactions, particularly in scenarios such as pedestrian-AV interactions. Further research will focus on systematically developing a decision-making model that describes decision behaviors, grounded in an understanding of the cognitive mechanisms underlying human decision uncertainty in emergency situations. By integrating cognitive principles with engineering methodologies, such a model is expected to align AV technologies with human behavioral patterns and needs, ultimately enhancing the safety and efficacy of these interactions.

Nevertheless, it is essential to recognize the limitations of the present study. First, while the validity of decision-making distributions observed in simplified experiments was partially supported by VR experiments, notable differences remain between the third-person perspective participants experienced in these experiments and the first-person perspective that pedestrians face in real-world traffic scenarios. To address this discrepancy, future work, including our subsequent studies, need to employ multi-level experimental designs—such as incorporating visual elements—to more accurately capture how pedestrians perceive their environment and make decisions in actual traffic environments. These efforts would enhance the

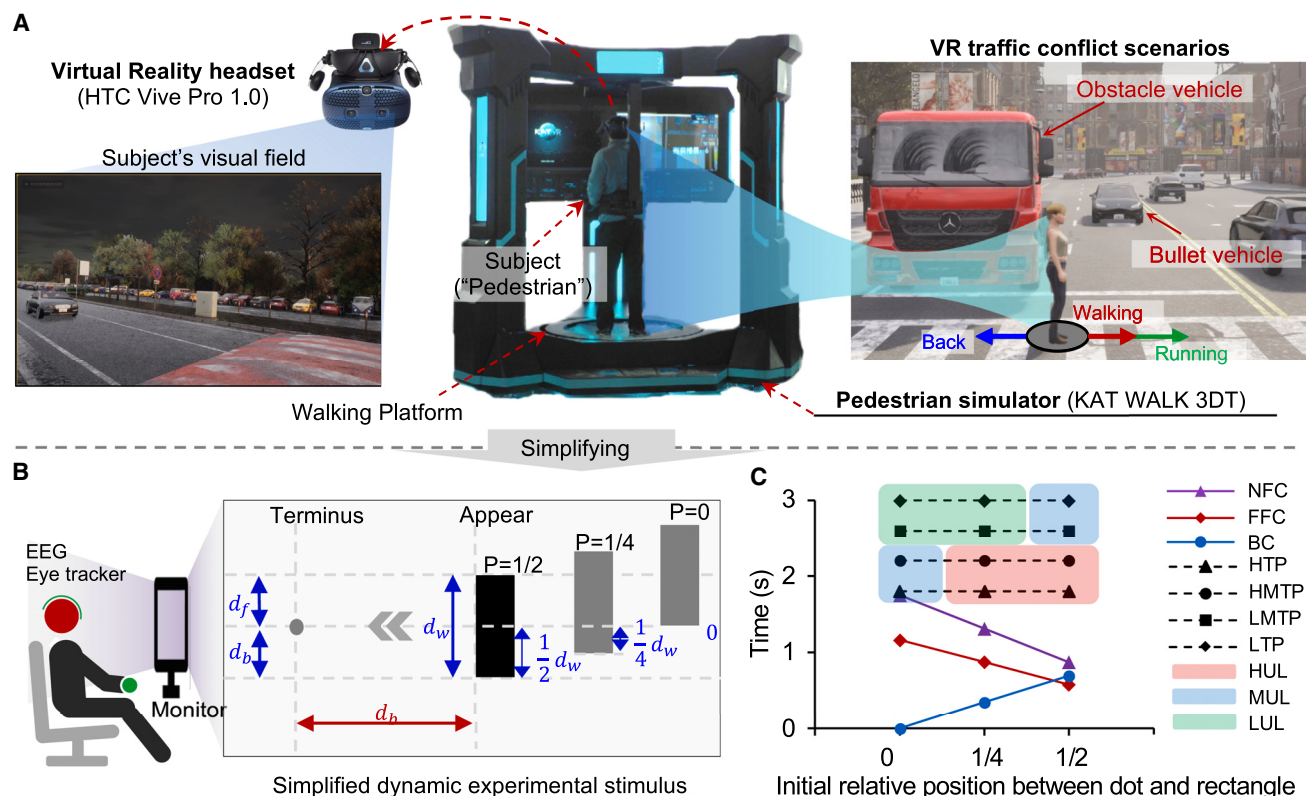


Figure 7. Experimental design

(A) A virtual-reality-based platform for studying pedestrian decision-making behavior.

(B) A simplified dynamic stimulus experiment. The initial relative position between the point and the rectangular block at the beginning of the experiment is shown.

(C) Design of experimental parameters of the simplified dynamic stimulus experiment. Dashed curves labeled “NFC” and “FFC” were used to depict the time taken for the dot to reach the upper end of the rectangle at normal and faster velocities, respectively, while the dashed curves labeled “BC” represent the time taken for the dot to reach the bottom end of the rectangle at normal and faster velocities. The solid curves denoted as “HTP” (high time pressure), “HMTP” (high-moderate time pressure), “LMTP” (low-moderate time pressure), and “LTP” (low time pressure) were utilized to represent four levels of time pressure ($t_0 = 1.8, 2.2, 2.6, \text{ and } 3.0 \text{ s}$), which indicates the time it takes for the rectangle to reach the dot. The transparent region encompasses conditions categorized into three urgency levels: high (HUL), moderate (MUL), and low (LUL). This classification integrates considerations of both initial position and time pressure.

experimental framework and contribute to a more comprehensive understanding of pedestrian decision-making mechanisms in realistic, hazardous traffic situations. Second, our analysis did not incorporate demographic factors including age and gender, which may restrict the applicability of the findings across diverse populations. Furthermore, the experimental stimuli were primarily designed to simulate urgent scenarios, thereby excluding a broader range of non-urgent conditions. Future studies should expand the variety of stimuli and participant groups to facilitate a more detailed discussion and broader generalization of the results.

METHODS

Participants

We recruited twenty-four healthy participants (seventeen males, mean age $\pm$ SD = 23.5 ± 2.6 years, range 18–27 years old) from Tsinghua University, China, through online recruitment. Of these participants, seventeen were involved in a simplified dynamic stimulus experiment that focused on collecting large-scale

experimental data, totaling 2,570 cases, and extracting multidimensional physiological signals. The remaining seven participants took part in a VR experiment designed to validate findings under comparable urgency levels, contributing an additional 448 cases to the study. One participant from the simplified dynamic stimulus experiment who did not complete the required experiment was excluded from the study. Detailed information about the participants is provided in Table S1. All participants were right-handed, had normal or corrected-to-normal vision, and had no history of psychiatric illness or head injury. The experimental procedures were approved by the Institutional Review Board (IRB) of Tsinghua University, and each participant provided informed consent before participating in the experiments.

General task procedures of the VR-based pedestrian-vehicle interaction experiment

The experiment was conducted using a pedestrian walking platform (KAT WALK 3DT), which allows participants to simulate positional changes within a virtual environment while walking in place (Figure 7A). Participants were equipped with VR headsets

(HTC Vive Pro 1.0) and immersed in traffic scenarios. The pedestrian walking platform can perceive the participants' positions in real time and replicate these positions within the virtual environment.

This experiment designed a virtual traffic scenario at an intersection, featuring a realistic road and building environment, randomly generated traffic flow, a “bullet vehicle,” and an “obstacle vehicle” for making a pedestrian-vehicle conflict (developed using Python 3.8 on the Carla 0.9.15 platform). Participants (pedestrians) wore VR headsets and entered the walking platform as pedestrians, as illustrated in Figure 7A.

At the beginning of the experiment, the participant appeared on one side of the crosswalk and could freely choose when to cross based on the traffic signals. As the pedestrian approached the front of the obstacle vehicle, a bullet vehicle would randomly appear on either the left or right side of the pedestrian and drive toward them. In all trials, the initial distance between the pedestrian and the bullet vehicle was consistently set to 20 m. The velocity of the bullet vehicle was set at 15, 25, 35, and 45 km/h. Correspondingly, the time pressures were calculated into four levels: 4.8, 2.9, 2.1, and 1.6 s. It is important to note that while the experiment was designed with four levels of TTC values and fixed initial positions for pedestrians and the bullet vehicle, the moment when pedestrians notice the approaching bullet vehicle is random. Consequently, the relative position between the pedestrian and the vehicle when the pedestrian observes the bullet vehicle and decides is also random, as illustrated in Figure S2. During the experiment, pedestrians walked at an average normal speed of 1.4 m/s. Upon detecting danger, they could either run or retreat (including stopping), with the running speed set at 2 m/s and the retreating speed at -1.4 m/s. The pedestrian platform could sense changes in behavior in real time based on gait variations. A trial was considered valid if the pedestrian either safely crossed the road or collided with a vehicle.

Before the experiment began, participants were given 3–5 min to familiarize themselves with the virtual environment and the pedestrian walking platform. During the formal experiment, each participant was required to complete 64 valid trials, divided into two sessions of approximately 2 h each, with a 5–10 min break between sessions. We recorded the relative position between the pedestrian and the bullet vehicle, the TTC, and the pedestrian's subsequent decision-making behavior upon observing the bullet vehicle through VR glasses. Pedestrian decision-making behaviors were categorized into three types: walking, running, and retreating (including stopping). Upon completion, each participant received a compensation of \$56.60. A total of 448 valid VR experiments were conducted.

General task procedures of the simplified dynamic stimulus experiment

The experiment was performed in a dimly lit and quiet room, utilizing a 27 inch monitor with a resolution of $2,560 \times 1,440$ and a refresh rate of 60 Hz. Participants were seated with their chin resting at a distance of 50–55 cm from the monitor.

The decision-making interactions between pedestrians and vehicles are complex in real traffic scenarios, potentially involving hesitancy or varying speeds of movement. We posit

that pedestrians' hesitancy and changes in speed fundamentally reflect the continuous decision-making process they undergo when interacting with vehicles.⁴⁸ Each trial in this study enabled participants to make a single-stage decision, mimicking and breaking down the continuous decision-making behavior of pedestrians. Surrounding vehicles, environmental complexity, and auditory stimuli in real traffic contexts can also influence pedestrian decision-making behavior. Yet, real-world accidents mainly involve single-pedestrian and single-vehicle incidents, as evidenced in statistics.⁴⁹ To reasonably simplify the pedestrian-vehicle interaction process into a well-defined framework, our study employed two-dimensional (2D) visual stimuli. This approach allowed us to focus on the key variables influencing pedestrian decision-making and facilitated a detailed examination of the underlying mechanisms.

The experimental stimulus, which consisted of a gray dot representing a pedestrian with a radius of 1.5° and a black rectangle symbolizing a vehicle with a width of 9.8° , was developed using the PsychoPy package in PyCharm and is illustrated in Figure 1B. The initial position of the gray dot was randomly displayed on the left or right side of the screen, positioned at the center of the screen vertically. The black rectangle appeared randomly at one of three prescribed initial vertical positions (see Figure 7). When the dot appeared on the left, the rectangle appeared on the right, and vice versa. The initial distance (d_b) between the dot and the rectangle was consistently set at 20.5° in all trials.

At the beginning of the experiment, the gray dot first appeared on the screen. After a random time interval ranging from 7 to 10 s, the black rectangle would then appear and crash into the dot (see Figure 1B). The velocity of the rectangle (v_o) was determined by the specified time pressure ($v_o = d_b/t_o$). To induce a sense of urgency, four levels of time pressure ($t_o = 1.8, 2.2, 2.6$, and 3.0 s) were designed based on extensive experience from numerous preliminary experiments (see Figure 6B). Participants controlled the direction and speed of the dot by pressing buttons to avoid the rectangle. Pressing button “1” caused the dot to move upward at a normal velocity (NFC: $9.8^\circ/\text{s}$), simulating pedestrians crossing the road while walking. Pressing button “2” made the dot move upward at a faster velocity (FFC: $13.5^\circ/\text{s}$), simulating pedestrians running across the road. Pressing button “3” led to the dot moving downward at a normal velocity (BC: $-9.8^\circ/\text{s}$), simulating pedestrians moving backward to avoid the obstacle (see Figure 6B). Subsequently, both the dot and the rectangle came to a halt when the rectangle reached the vertical position of the dot. The screen displayed the score for the current trial, as shown in the decision feedback, as well as the total score for the ongoing group. Finally, the screen transitioned to a blank page, and the gray dot reappeared.

Participants were provided with a clear description of the environment prior to the experiment. They were first instructed to perceive the black rectangle as an approaching vehicle and the gray dot as the pedestrian. This setup engaged the participants in a conflict situation where a collision would result in a severe penalty, representing themselves in the physical scene. This method aimed to mitigate the impact of first-person and third-person perspectives on their decision-making processes. This initial guidance aimed to facilitate the mental construction of the experimental scenario. Subsequently, participants

underwent training to familiarize themselves with the experimental procedures and gain essential preliminary knowledge. Successful completion of the training was determined by attaining a total score of 3,000 points. Furthermore, the formal experiments commenced, with each session lasting approximately 20 min per group. Finally, post-experiment feedback from participants confirmed the effectiveness of our experimental design in accurately replicating pedestrian decision-making behaviors when interacting with vehicles, as it successfully simulated the tense situations that arise during such interactions.

The reward rules in the experiment were designed to incentivize participants to successfully navigate the traversal task. Participants scored 100 points if the participant-controlled dot avoided colliding with the rectangle by NFC behavior and 40 points if the participant-controlled dot avoided colliding with the rectangle by FFC behavior. This reflects that pedestrians running through requires more effort compared to walking, hence the reduction in the reward. However, no points were awarded if participants avoided colliding by BC behavior, symbolizing pedestrians not completing the traversal task, for which there is no reward. These reward rules were established based on extensive preliminary experiments and feedback from participants. Additionally, an infinitely high penalty was implemented to simulate the consequences of traffic accidents, resulting in the total score being reset in the event of a collision.

In order to incentivize participants to approach each trial with the necessary level of seriousness, they received a fixed remuneration of \$36 as well as an additional reward (\$0–\$14) based on the number of trials completed. Therefore, participants who completed the experiment with fewer trials were eligible for higher rewards, ensuring that they were motivated to strive for efficiency and accuracy throughout the experimental process.

During the formal experiment, each participant was required to conduct 3–5 sets of experiments. Within each set, the experiment terminated automatically once the participant attained a score of 3,000 points or encountered a collision. Any set in which the participant scored above 1500 points was deemed valid experimental data. Moreover, the experiment ceased when the cumulative scores from all sets collectively exceeded 6,500 points.

Data acquisition and processing

The decision-making behavior of participants involves the avoidance actions (Be = (NFC, FFC, BC)) selected in each trial, with the response time defined as the interval between the onset of the black rectangle and the initiation of the keystroke. The uncertainty of participant decisions is quantified using entropy (Equation 1), and it is observed that participants' decision-making behavior may exhibit variability across multiple trials under identical stimulus conditions.

$$\text{Entropy} = - \sum_{x \in \text{Be}} p(x) \log p(x) \quad (\text{Equation 1})$$

The continuous EEG recordings were obtained using the Curry 8.5 system and the SynAmps RT 64-channel amplifier from Neuroscan. Electrode placement followed the international 10/10 system³⁷ (refer to Figure S1), and data were digitized at a rate of 2,000 Hz.⁵⁰ Electro-oculogram (EOG) signals were captured

from two electrodes placed above and below the left eye, as well as two electrodes positioned at the outer canthus of each eye. Subsequent EEG data processing was carried out in MATLAB 2019, employing custom scripts and routines from EEGLAB v.2021.1.

The preprocessing steps comprised the following procedures: (1) resampling the sampling frequency to 500 Hz; (2) applying a 0.5–45 Hz band-pass filter to the data using a Hamming filter; (3) identifying and interpolating noisy channels based on elevated signal variance using a spherical spline; (4) re-referencing each channel signal to the average of all electrodes; (5) extracting epochs of stimulus time-locked data from continuous EEG recordings within the range of –1,000 to 1500 ms, followed by baseline correction with respect to –500 to 0 ms; (6) extracting epochs time locked to response moments within the range of –1,500 to 500 ms; (7) employing the RunICA algorithm for component decomposition to remove signal artifacts stemming from blinking, glancing, heartbeat, movement, etc.; (8) conducting amplitude checks on all epochs and rejecting those with amplitudes exceeding 100 μV ; and (9) implementing current source density (CSD) transformation⁵¹ to reduce the impact of spatial location on event-related potential components. Subsequently, the epochs were averaged to generate an average waveform for different experimental conditions and each subject.

We recorded EEG components from various regions of the brain, including P1, N2c, P3a, P3b, MRPs, CPP, and FCN, with measurements acquired at OZ, P7/P8, FZ, PZ, C3, and fronto-central midline (FCz), respectively. The latencies of P1, N2c, P3a, and P3b were determined based on the moment when the amplitude reached a positive or negative peak in the stimulus-locked waveform. In addition, the amplitude peaks of all components were determined by calculating the average amplitude within a 100 ms window centered around the peak moment in the continuous waveform. Furthermore, the peak time (relative response) of MRPs, CPP, and FCN was measured at the moment when the amplitude reached its peak in the response-locked waveform. The pre-response amplitudes of CPP and FCN were quantified as the average value of the amplitude within a 50 ms window before the response. The build-up rate of CPP and FCN was assessed as the slope, identified by linearly fitting the waveform between –150 and –50 ms relative to the response.

The pupil diameter was meticulously measured using a Tobii Pro Spectrum eye tracker from Sweden at a sampling rate of 1,200 Hz. Data processing involved the use of the Gaze Filter (Tobii I-VT-Fixation) within Tobii Pro Lab software and custom scripts developed in MATLAB 2019. The recorded data's validation accuracy and precision were 0.3 ± 0.01 and 0.16 ± 0.01 , respectively. Furthermore, the sampling frequency was re-sampled to 300 Hz.

Statistical analysis

We first performed one-way/two-way ANOVA and calculated the Pearson correlation coefficient to evaluate differences and associations in pedestrian decision-making behavior (response time, uncertainty) at different urgency levels, given that the behavioral data samples exhibited a normal distribution. Additionally, we used the Friedman test to examine the significant impact of

urgency levels on neurophysiological signals. Furthermore, the correlation between neurophysiological signal features and response time, as well as decision uncertainty, was investigated using the Spearman correlation test. All statistical analyses were conducted as two-tailed tests, and in non-parametric tests, data falling within the 95% confidence interval were retained to eliminate outliers. Please refer to the [Note S1](#) section in the [supplemental information](#) for detailed calculation formulas of the statistical analysis methods.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for data and code should be directed to and will be fulfilled by the lead contact, Bingbing Nie (nbb@tsinghua.edu.cn).

Materials availability

This study did not generate new unique materials.

Data and code availability

- All data reported in this paper will be shared by the lead contact upon request.
- All original code has been deposited in Zenodo (<https://zenodo.org/records/14257262>) and is publicly accessible as of the publication date.
- Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

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AUTHOR CONTRIBUTIONS

B.N. coordinated the working project; Q.L. and B.N. designed the experiments and wrote the paper; Q.L. performed research; Q.L., S.L., and S.S. analyzed data; and B.L., X.G., and J.W. contributed to the discussion on experiment design and data analysis.

DECLARATION OF INTERESTS

The authors declare no competing interests.

SUPPLEMENTAL INFORMATION

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